

Human Emotion Recognition by Fuzzy Spatial-Temporal Graph Convolution through Dynamic Action Videos

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Abstract

The research investigates action-based emotion recognition, an underexplored domain compared to facial expression analysis. It uses skeleton sequences for their computational efficiency and intrinsic graph-structured representation. In contrast to prevailing approaches that employ either oversimplified discrete emotion categories or highly specialized valence-arousal dimensional representations, it incorporates a comprehensive 29-category emotion taxonomy to achieve finer-grained classification. The technical contribution includes a novel fuzzy transformation mechanism that bridges dimensional and categorical emotion spaces through membership functions, thereby facilitating the utilization of existing annotated datasets while maintaining interpretability. Furthermore, it enhances classification robustness by incorporating fuzzy set theory. Experiments substantiate the efficacy of the approach. The code is publicly available on GitHub: <https://github.com/YeeHoran/Fuzzy-Human-Action-based-Emotion-Recognition>.

CCS Concepts

• Human-centered computing~Human computer interaction (HCI) • Computing methodologies~Artificial intelligence~Computer vision~Computer vision tasks

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Keywords

Human Computer Interaction, Emotion Recognition, Graph Convolution, Fuzzy Systems

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1 Introduction

The concept of "enabling machines to recognize emotions" was introduced by Marvin Minsky at MIT^[1], who emphasized emotions' role in achieving computer intelligence. Humans primarily acquire information through vision (83%), with smaller contributions from hearing (11%), smell (3.5%), touch (1.5%), and taste (1%)^{[2][3][4][5][6]}. While emotion recognition has mainly focused on facial expressions, other visual cues such as body poses and actions are also crucial for detecting emotions. These cues are harder to fake, remain visible even when faces are obscured, and can convey different emotional states when combined with facial expressions. Therefore, emotion recognition should also incorporate body gestures and behaviors.

Body gestures and behavior-based recognition commonly extract skeleton key points from videos. Although research in action skeleton-based emotion recognition is underexplored, there is extensive research in macro and micro worlds through graph analysis^{[7][8][9][10]}. In the macro fields, it is used in social networks^[11], traffic networks^[12]. In micro worlds, it is applied to biological protein molecule prediction, drug molecule structure detection, and chemical material molecular modeling. Therefore, spatial-temporal graph models could be similarly used to simulate human action sequences where the key points in each frame are nodes in a graph^[13] and the connections between key points within the frames as well as those connecting adjacent frames with the same key points

are the edges of the graph. Thus, the study will employ the body action series graph to recognize emotion categories.

Most emotion datasets use standard 7 categories (Happy, Angry, Sad, etc.)^[14], but they are sparse and lack comprehensive coverage. Valence-Arousal annotations provide finer emotional parameters but are less intuitive and harder to perceive. To address this, fuzzy membership functions^[15] can map precise Valence-Arousal values to broader emotion categories^[16] as shown in Fig.1 and Table 1, and will be more applicable for richer emotion term scenarios.

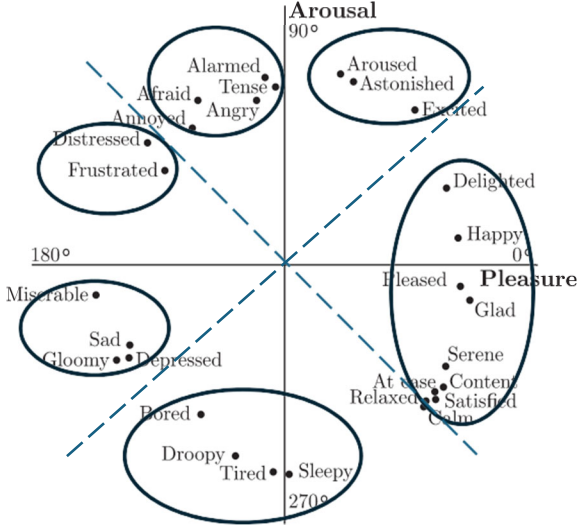


Figure 1: This is 28 emotion categories^[16] & 6 regions in Valence-Arousal space.

Additionally, through observing the emotion categories for action video samples dataset with broader range of emotional category labels (Fig.1^[16]), it is obvious that some emotion concepts are very similar to each other. For example, Angry and Annoyed, Distressed and Sad, Content and Satisfied, Delighted and Happy are very alike to each other, which is consistent with their presented coordinates as well, which is distinctly clustered in Fig.1. Therefore, the fuzzy relationships^[17] between fuzzy emotional sets can be introduced to emotion recognition models to fully utilize the fuzziness of emotion category terms for constructing an emotion recognition model with higher performance. The main contributions of this research are threefold: 1. Recognizing emotions through dynamic action behavior sequences, aiming to include broader emotion-related data sources into the field of affective computing. 2. Establishing a more comprehensive emotion labeling method by fuzzy set membership functions. It provides a way to label datasets with Valence-Arousal parameters to generate more comprehensive emotion labels. 3. Utilizing the fuzzy similarity relationships between fuzzy emotion sets to build a better optimization model for predicting emotions.

2 Preliminary Definitions

2.1 Fuzzy Membership Function^[18]

The whole field of emotion in Fig. 1 is noted by U , as shown in (1), where V represents Valence (Pleasure), and A represents Arousal.

Each point u_{VA} in field U is confined by (2). Each emotion category is noted as a fuzzy subset E_i ($i = 0, 1, 2, \dots, 28$). Their valence and arousal values are quantified in Table 1.

$$U(u_{VA} \in U) \quad (1)$$

$$u_{VA}(V \in [-10, +10], A[-10, +10]) \quad (2)$$

Table 1: Valence-Arousal for 29 Emotions

Emotion Category Label	Valence	Arousal
Alarmed	-0.9	8.6
Afraid	-4.1	7.6
Annoyed	-4.36	6.3
Distressed	-6.52	5.7
Frustrated	-5.66	4.3
Tense	-0.4	8.2
Angry	-1.3	7.6
Miserable	-8.92	-1.4
Sad	-7.34	-3.7
Gloomy	-7.98	-4.4
Depressed	-7.38	-4.3
Bored	-3.92	-6.9
Droopy	-2.3	-8.8
Tired	-0.5	-9.6
Sleepy	0.24	-9.6
Relaxed	6.72	-6.6
Calm	6.6	-6.6
Serene	7.66	-0.48
Content	7.56	-5.6
Satisfied	7.18	-6.2
At ease	7.16	-5.8
Pleased	8.34	-1
Glad	8.8	-1.6
Happy	8.2	1.2
Delighted	6.66	3.6
Aroused	2.7	8.8
Astonished	3.2	8.4
Excited	6.2	7.2
Neutral	0	0

For each emotion E_i , its fuzzy subset is defined in (3).

$$E_i(c_{E_i} \in E_i, (u_v - c_v)^2 + (u_a - c_a)^2 \leq r^2) \quad (3)$$

Where c_{E_i} is the center of fuzzy emotion subset E_i , its coordinates (c_v, c_a) are the Valence-Arousal values in Table 1, and r represents the radius of fuzzy emotion subset E_i .

The fuzzy membership function for each fuzzy emotion subset E_i ($i = 0, 1, 2, \dots, 28$) is defined as $\tilde{E}_i(u)$ in (4), which is Gaussian membership function to guarantee smooth transition. It is based on the distance of a sample u_{VA} in U to the center of fuzzy emotion subset E_i , normalized by radius r . Where $d(u, c)$ is the distance between u and c ; $u \in U$, c is the center of E_i .

$$\tilde{E}_i(u) = \begin{cases} \exp\left(-\left(\frac{d(u, c)}{r}\right)^2\right), & d(u, c) \leq r; \\ 0, & d(u, c) > r. \end{cases} \quad (4)$$

2.2 Fuzzification by Fuzzy Priority Sorting^[19]

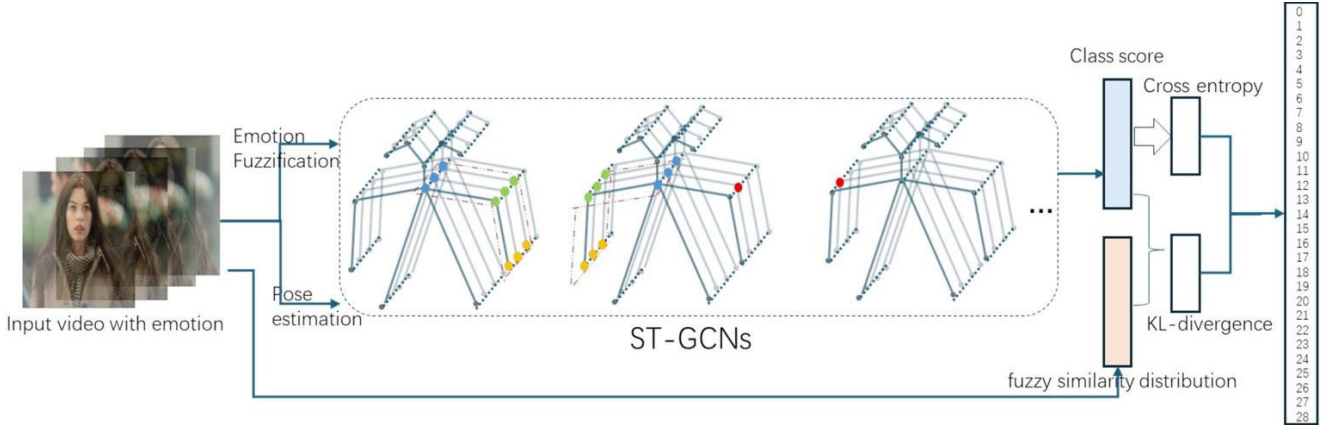


Figure 3: Fuzzy Emotion Recognition by ST-GCN.

The 29 emotion category terms like Aroused, Astonished, Happy, etc. are fuzzy concepts, while the valence-arousal values are precise values. In the study, it uses membership sorting for input data fuzzification to obtain emotion categories from valence-arousal values. This is, after all emotion memberships of 29 $\tilde{E}_i(u)$ ($i = 0, 1, 2, \dots, 28$) for u_{VA} are computed, it sorts them in descending order, as shown in (5).

$$(\tilde{E}_1^*(u), \tilde{E}_2^*(u), \dots, \tilde{E}_n^*(u)), \tilde{E}_1^*(u) \geq \tilde{E}_2^*(u) \geq \dots \geq \tilde{E}_n^*(u) \quad (5)$$

Then, the emotion category term with the highest membership is selected as the corresponding emotion category for u_{VA} , as formulated in (6).

$$E^* = \arg \max_{E_i} \tilde{E}_i(u) \quad (6)$$

2.3 Fuzzy Relationships^[20] Among Fuzzy Emotion Subsets

The fuzzy relationship between two fuzzy set E_i and E_j is $\tilde{R}(E_i, E_j)$, where i is fixed, and j ranges from 0 to 28. Thus, it is a vector representing the emotion similarity between E_i and all 29 emotion categories.

Each of the four quadrants could be divided by 45-degree lines, resulting in 6 major emotion clusters, with strong intra-cluster similarity, as illustrated in Fig.1. Therefore, if two emotion categories fall to different clusters, they obviously belong to different emotional classes. Based on this observation, using cosine angular similarity is more meaningful than traditional Euclidean distance, Manhattan distance, etc., since emotion's orientation angle is more crucial for distinguishing them. Thus, the fuzzy similarity relationship between any two emotions could be calculated by (7).

$$\cos(E_i, E_j) = 1 - \frac{E_i \cdot E_j}{\|E_i\| \|E_j\|} \quad (7)$$

Where E_i and E_j are two emotion fuzzy sets, $\cdot$ is dot product, and $\|$ is Euclidean norms of $E_i(V, A)$ and $E_j(V, A)$. It is calculated by cosine distance since it could be observed that the circumflex emotion model (Fig.1) could be generally divided into 6 major clusters.

3 Method

3.1 Spatiotemporal Graph of Action Series

It uses coco 17 key points skeleton to represent human's skeleton structure in the research, as shown in Fig. 2.

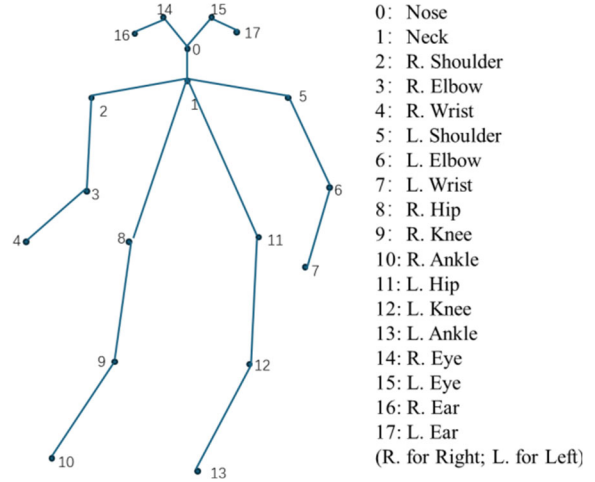


Figure 2: Key points for human poses according to the COCO output format.

It could be observed that besides body key points, it contains 5 key points on faces as well. They are left ear, right ear, left eye, right eye, and nose, which are useful for recognizing emotion conditions. With such moving postures and actions, it provides more comprehensive emotion information which could be extracted as emotion features.

3.2 Scheme Structure

The fuzzy spatial-temporal graph convolution action-based emotion recognition scheme is illustrated in Fig.3. There are mainly two parts consist of preparation phase and model learning phase.

In preparation phase, emotion fuzzification is performed to transform valence-arousal inputs from dynamic action emotion

dataset to one of the 29 emotion category labels by fuzzy membership function and fuzzy priority sorting formulated in Section 2. Then, fuzzy set similarity relationships^[21] are calculated according to their similarity in emotion orientation which is set up in Section 2 as well. Pose recognition^[22] is carried out on dynamic action videos to produce spatial-temporal graph data as the input to the second phase. These steps are illustrated in left part of Fig. 3.

In the second model learning phase, first, fuzzy emotion annotations with spatial-temporal graph data are fed into the spatial-temporal graph convolution network as shown in Fig.3. They are processed by spatial-temporal graph convolution to generate output emotion category distribution, which is calculated by distribution alignment with fuzzy similarity relationship distribution vector, to generate Kullback-Leibler divergence loss, which will be presented below. This Kullback-Leibler divergence loss and normal cross-entropy loss will be combined to make final decision for emotion category prediction.

3.3 Kullback-Leibler Divergence Constrained Optimization

To compute the similarity between two probability distributions for output class scores and fuzzy emotion similarities, it uses Kullback-Leibler Divergence^[23]. It measures how one probability distribution diverges from another expected probability distribution. It is often used to compare two distributions and is defined in (8).

$$D_{KL}(O||S) = \sum_k O(u) \log \frac{O(u)}{S(u)} \quad (8)$$

Where $O(u)$ represents the output class score for data sample u , and $S(u)$ represents the fuzzy similarity relationship vector of predicted emotion for sample u . The value of Kullback-Leibler divergence is non-negative, and the larger the value, the more different the two distributions are.

Therefore, the final optimization with fuzzy emotion similarity relationship constraints is defined in (9), where $L_{cross-entropy}$ is the normal loss, and $L_{KL-divergence}$ is computed by (8).

$$E_i = \underset{E_i(i=0,1,\dots,28)}{\operatorname{argmax}} (L_{cross-entropy} + \lambda L_{KL-divergence}) \quad (9)$$

Where λ modulates the weights of the two losses.

4 Simulation Results

4.1 Dataset

The research uses AFEW-VA dataset^[24] that is a collection of highly accurate per-frame annotations levels of valence and arousal for 600 challenging video clips extracted from feature films.

4.2 Setup

The simulation procedures are executed on an integrated development environment Pycharm2024.2.3, an Intel(R) Xeon(R) E5-2678 v3 CPU operating at 2.50GHz with 64.0GB RAM, and windows10 operating system. To facilitate deep learning, it employs NVIDIA RTX3090ti GPU, CUDA 11.8 and CuDNN9.1.0 tools.

4.3 Simulation Results

It executed ablation test on the dataset by original spatial temporal graph convolution, and the proposed fuzzy set system enhanced spatial temporal graph convolution.

The batch size is set to 16, the training epoch number is 64. Under the setup, the simulation results are presented in Table 2.

Table 2: Ablation Test Results.

Method	top1_acc	top5_acc
Original STGCN	0.3438	0.75
proposed	0.40625	0.78125

From Table 1, it shows that the proposed fuzzy set enhanced spatial-temporal graph convolution action-based emotion recognition obtains better performance than the original method. Thus, it could be a dynamic action-based emotion recognition baseline for further research on action-based emotion recognition.

5 Conclusions

The study explores emotion recognition based on dynamic action video sequences, aiming to introduce a broader range of emotion expression modalities into affective computing. Additionally, a more extensive set of emotion categories is used to meet the growing demand for perceiving nuanced emotional classes. Since emotion category terms are inherently fuzzy concepts, and different emotion subsets have overlapping boundaries, it proposes a discrete emotion annotation method based on fuzzy membership functions and fuzzy ranking for datasets containing continuous Valence-Arousal emotion parameters. Furthermore, by analyzing the spatial distribution of 29 emotion categories in the Valence-Arousal space, it finds that the emotion label set can be roughly clustered into six groups, where intra-cluster emotions are highly correlated while inter-cluster correlations are low. Based on this observation, it builds a fuzzy similarity relationship set for the 29-emotion fuzzy set. The set was then incorporated to optimize the model to enhance emotion recognition accuracy. Simulation experiments demonstrate that using membership function-based emotion annotation and fuzzy similarity constraint-based optimization model improves recognition accuracy. Additionally, the study provides a baseline for emotion recognition from action videos.

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